

Navigating Waste: A Comprehensive Analysis of Waste Treatment Practices in Canada, the US, China and Australia

Abstract:

Waste treatment is a pivotal aspect of global environmental sustainability, and this study provides a comprehensive examination of waste management practices in Canada, the United States, China, and Australia. Focusing on Canada, the analysis spans municipal solid waste landfills, operational trends, and governance structures from 2002 to 2020. Noteworthy reductions in waste per person amid increasing total waste production reveal positive environmental shifts, possibly influenced by public awareness and recycling initiatives. The federal, provincial/territorial, and municipal roles in waste management governance are explored.

Comparative analyses with the United States, China, and Australia offer insights into historical trends, challenges, and sustainable strategies. Despite progress in recycling initiatives, the United States continues to grapple with high landfill usage, emphasizing the persistent challenge of reducing landfill dependence. China's waste management journey reflects economic development, with a comparatively moderate increase in per-person waste generation. Australia's waste composition data underscores the significance of targeted initiatives to address specific materials in the waste stream.

This study provides a nuanced understanding of waste management complexities, progress, and challenges, highlighting the importance of sustained global efforts to promote circular waste management systems and environmental sustainability.

I. Introduction

Waste management plays a pivotal role in shaping the environmental and social landscape of nations. In Canada, the vast expanse of over 3,000 municipal solid waste landfills forms a critical part of the waste disposal infrastructure. This overview delves into the intricate details of these landfills, shedding light on their operational status and their substantial contribution to Canada's waste disposal. Examining the trajectory of Canada's waste management progress from 2002 to 2020 reveals a nuanced story. Despite a surge in total waste production, a commendable reduction in waste per person suggests positive environmental shifts, possibly fueled by public awareness and recycling initiatives. Understanding the governance structure is imperative, and this section navigates through the roles of the federal, provincial/territorial, and municipal governments in overseeing waste management. An analysis of tonne/person ratios across provinces uncovers intriguing variations, with Alberta consistently leading in waste generation and Nova Scotia showcasing efficient waste management. Further exploration into waste management practices in the United States, China, and Australia offers a comparative lens, unravelling historical trends, challenges, and the pursuit of sustainable waste management strategies. Together, these sections provide a comprehensive overview of waste management,

encapsulating the complexities, progress, and challenges faced by nations in their pursuit of environmental sustainability.

II. Overview of Waste Management in Canada

According to the 2019 government statistics, there are over 3,000 municipal solid waste landfills across Canada. Slightly over half of the landfills remain active, while the remainder are closed. Among these, approximately 270 qualify as large landfills, defined either by a waste capacity exceeding 100,000 tonnes (in the case of open landfills) or by having accommodated more than 450,000 tonnes of waste (among closed landfills). Remarkably, these 270 landfills accounted for nearly 90% of the waste disposed of in landfills in 2019 and have collectively accepted over 70% of Canada's landfill waste since 1941 (Reducing Methane Emission, 2022).

Canada's Progress in Waste Management

Over the past two decades, Canada has witnessed notable progress in its waste management patterns. From 2002 to 2020, the total amount of waste produced increased, reaching a peak in 2006 at 26.4 million tonnes before stabilizing around 24-25 million tonnes in subsequent years. Concurrently, the Canadian population experienced steady growth, from 31.5 million in 2002 to 38 million in 2020. Despite the overall increase in waste, there is a positive environmental shift reflected in the decreasing trend of waste per person, dropping from 0.766 tonnes in 2002 to 0.687 tonnes in 2020. This decline suggests a commendable reduction in individual waste production, potentially attributed to heightened public awareness, recycling initiatives, and evolving environmental consciousness. The data underscores the complex interplay between population dynamics, environmental policies, and societal behaviours in shaping a nation's waste landscape, offering valuable insights for future sustainability endeavours.

Tonne Per Person In Canada

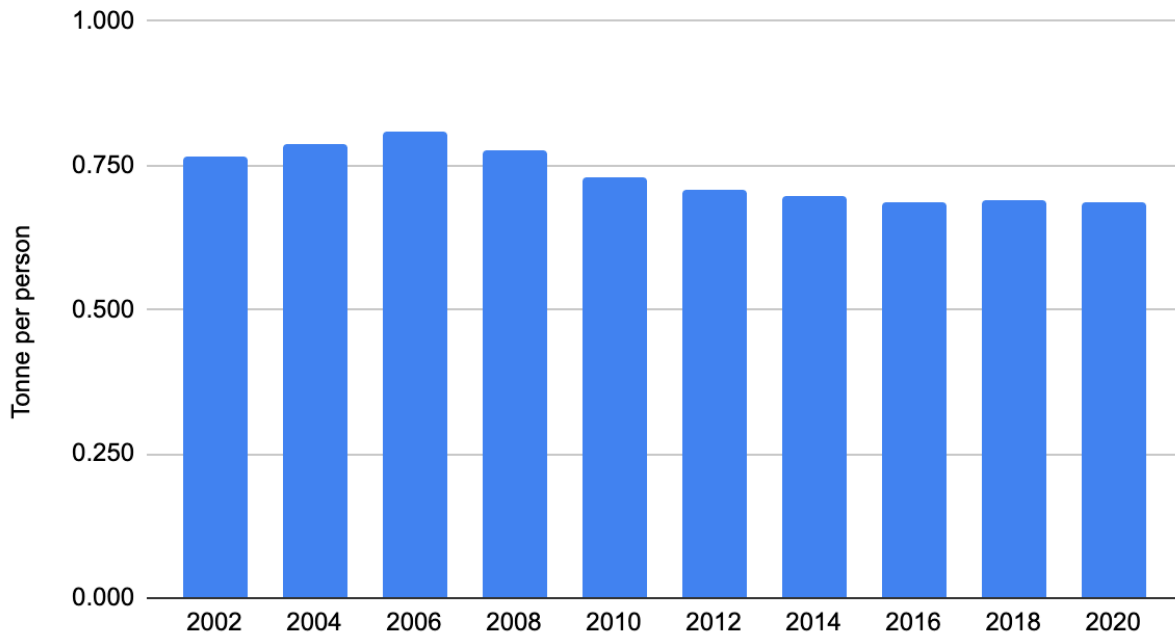


Figure 1: Waste (tonne) per person from 2002 - 2020 in Canada

Who Look Over Waste Management in Canada:

Waste Management in Canada is under the Canadian Council of Ministers of the Environment (CCME) environment ministers from the Federal, Provincial/ Territorial and Municipal governments. However, waste management primarily falls under the jurisdiction of provincial/ territorial and municipal governments.

Federal Government: The federal government in Canada is responsible for regulating hazardous waste management and international waste shipments. It oversees the transportation of hazardous materials, including hazardous waste, across provincial and international borders. Environment and Climate Change Canada is the federal agency responsible for regulating and enforcing environmental laws related to hazardous waste and pollution prevention.

Provincial and Territorial Governments: Each province and territory in Canada has its own regulations and policies governing waste management. They oversee the management and disposal of non-hazardous and household waste, as well as recycling programs. Provincial and territorial authorities often establish standards, issue permits for waste facilities and enforce waste management regulations.

Municipal Governments: Waste collection and disposal services, as well as recycling programs, are typically managed at the municipal level. Local municipalities are responsible for organizing waste collection, setting curbside pickup schedules, maintaining landfill and recycling facilities, and promoting waste reduction and recycling initiatives within their communities.

Provinces With Highest and Lowest Waste (tonne) per Person:

The analysis of the data reveals notable variations in the tonne/person ratio across provinces in Canada over the examined period. Alberta consistently stands out with the highest tonne/person values, peaking at 1.146 in 2008 and maintaining levels above 0.900 in subsequent years. This suggests a higher per capita waste generation in Alberta compared to other provinces, potentially influenced by economic factors or regional consumption patterns. On the opposite end of the spectrum, Nova Scotia consistently reports the lowest tonne/person ratios, with values dipping as low as 0.378 in 2008. This indicates a more efficient waste management or recycling system in Nova Scotia, resulting in lower per capita waste generation.

Highest Tonne/Person:	Province	Tonne/Person
2002	Alberta	0.919
2004	Alberta	0.944
2006	Alberta	1.107
2008	Alberta	1.146
2010	Alberta	1.045
2012	Alberta	1.004
2014	Alberta	0.999
2016	Alberta	1.000
2018	Alberta	0.955
2020	Alberta	0.917

Table 1: Province with Highest Tonne/Person 2002 - 2020

Lowest Tonne/Person:	Province	Tonne/Person
2002	Nova Scotia	0.416
2004	Nova Scotia	0.426
2006	Nova Scotia	0.383
2008	Nova Scotia	0.378
2010	Nova Scotia	0.389
2012	Nova Scotia	0.387
2014	Nova Scotia	0.388
2016	Nova Scotia	0.396
2018	Yukon, Northwest Territories and Nunavut	0.351
2020	Yukon, Northwest Territories and Nunavut	0.351

Table 2: Province with Highest Tonne/Person 2002 - 2020

United States

The waste management data from 1960 to 2018 illustrates a significant evolution in the United States. Over this period, the total waste generated increased from 88.1 million tonnes in 1960 to 292.4 million tonnes in 2018. Recycled waste, while showing a consistent upward trend, experienced a notable surge from 53.01 million tonnes in 2000 to 69.09 million tonnes in 2018. This suggests an increasing emphasis on recycling initiatives. Composting and combustion with energy recovery also witnessed growth, reflecting a broader adoption of environmentally friendly waste management practices. However, landfill usage remained consistently high, with 146.12 million tonnes in 2018, indicating ongoing reliance on traditional disposal methods. The introduction of other food management data in 2018, accounting for 17.71 million tonnes, highlights a focus on addressing food waste. While progress is evident in recycling and alternative waste management methods, the data underscores the continued challenge of reducing landfill dependence and curbing individual waste generation.

Municipal Solid Waste Management: 1960-2018

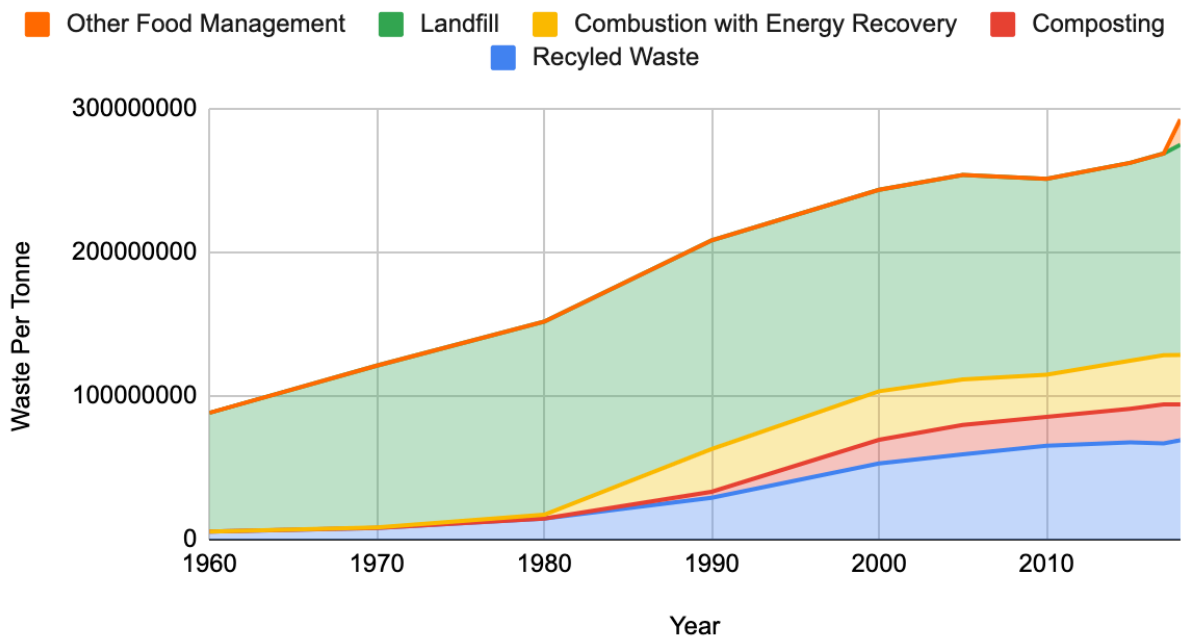


Figure 2: Different types of waste (tonne) generated from 1960 - 2010 in United States

The waste management data for the United States from 1960 to 2018 depicts a substantial increase in total waste generation over the examined period. The total waste grew from 88.12 million tonnes in 1960 to 292.36 million tonnes in 2018. This escalation can be attributed to several factors, including population growth and changes in consumption patterns driven by economic development and technological advancements. While the population nearly doubled during this timeframe, the waste per person also exhibited a noteworthy rise, increasing from 0.500 tonnes in 1960 to 0.880 tonnes in 2018. This suggests a concurrent trend of both individual and collective contributions to the growing waste stream. Efforts to address this surge in waste should focus on promoting sustainable practices, waste reduction, and recycling initiatives to mitigate the environmental impact associated with heightened waste generation. The stabilization of waste per person in the later years indicates a potential for improved waste management practices, yet continued efforts are essential to achieve more sustainable and circular waste management systems.

Waste (tonne) Per Person vs. Year

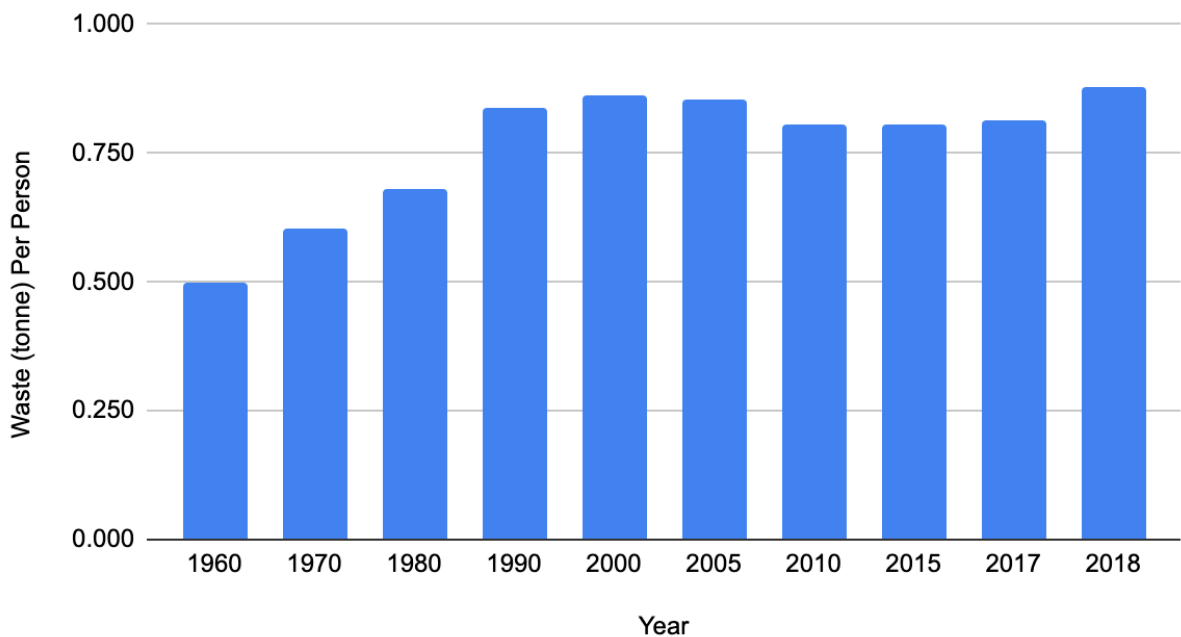
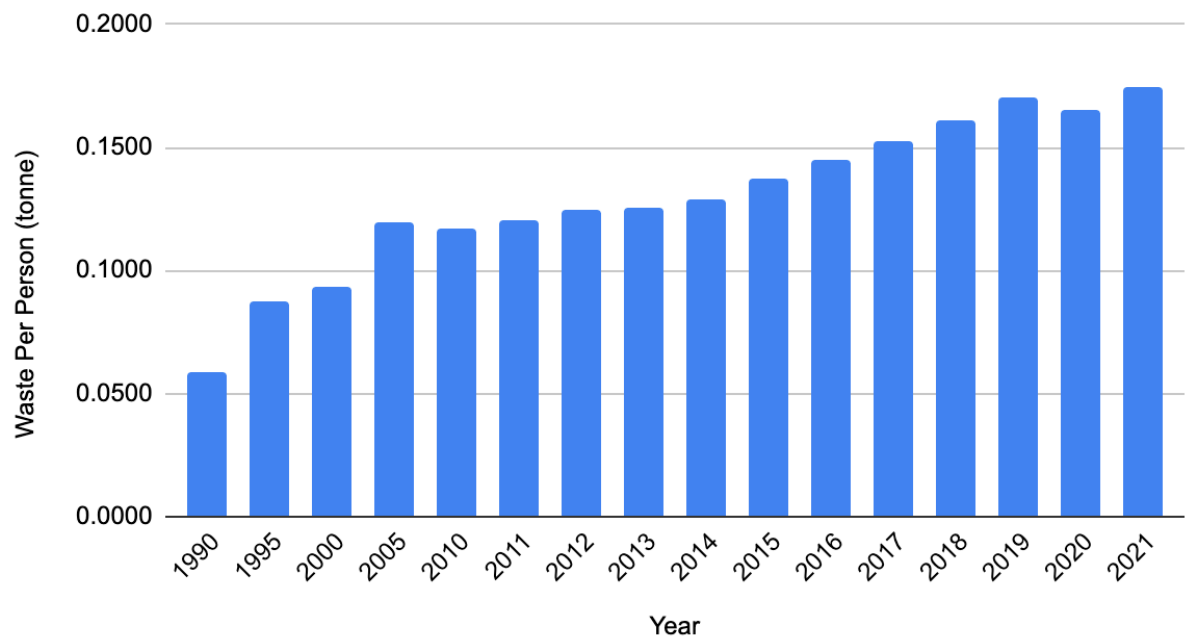


Figure 3: Waste (tonne) per person from 1960 - 2019 in United States

China:

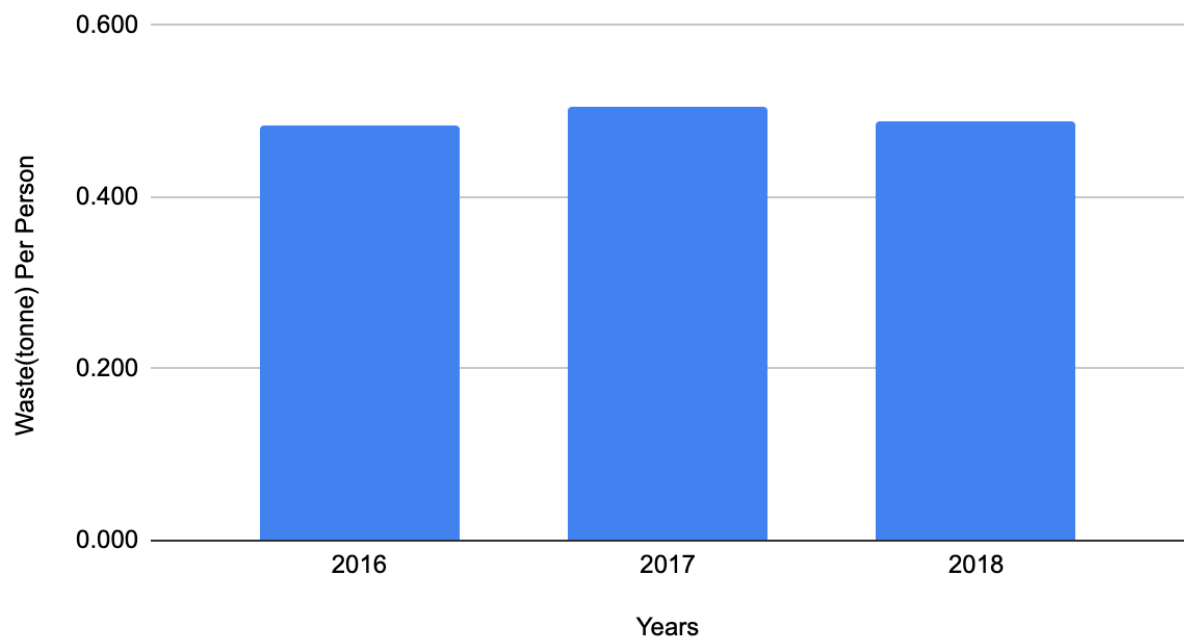
The waste management data for China from 1990 to 2021 demonstrates a substantial increase in both total waste generation and per-person waste production. The total waste generated has risen significantly from 67.67 million tons in 1990 to 248.69 million tons in 2021. This surge aligns with China's rapid economic development and urbanization during this period, leading to increased consumption and subsequently, waste generation. Despite the continuous growth in waste production, there is a noteworthy trend in the waste per person metric. Starting at 0.0587 tonne per person in 1990, it increased gradually, reaching 0.1744 tonne per person in 2021. This suggests a more moderate pace of per capita waste generation compared to the overall increase in waste, indicating potential improvements in waste management efficiency, recycling practices, and environmental awareness. However, the challenge remains to further enhance sustainable waste management strategies to address the environmental impact associated with the expanding waste stream.

Waste Per Person (tonne) vs. Year



The waste composition data for Australia from 2016 to 2018 offers insights into the nation's waste management landscape. Notably, organic waste consistently constitutes the largest share, peaking at 6.5 million tonnes in 2017, highlighting the significant contribution of biodegradable materials to the overall waste stream. Metals and paper & cardboard also represent substantial portions, showcasing the persistence of materials that are recyclable or, in the case of metals, recoverable. Plastics maintain a consistent presence, emphasizing the ongoing challenge of managing non-biodegradable materials. The waste per person metric, while fluctuating slightly, hovers around 0.49 to 0.51 tonnes per person, indicating a relatively stable individual waste generation pattern across the years. This data underscores the importance of targeted waste reduction and recycling initiatives, particularly in addressing the notable quantities of organics, metals, and plastics, to move towards a more sustainable and circular approach to waste management in Australia.

Waste (tonne) Per Person vs. Year



Conclusion:

In conclusion, this comprehensive exploration of waste management across Canada, the United States, China, and Australia provides a nuanced understanding of the complexities, progress, and challenges in each nation's approach to environmental sustainability. In Canada, the examination of over 3,000 municipal solid waste landfills highlights their significant role in waste disposal, while the analysis of the country's waste management progress underscores positive shifts in per capita waste production. Understanding the governance structure elucidates the roles of federal, provincial/territorial, and municipal authorities in overseeing waste management. Provincially, Alberta stands out with the highest waste generation, contrasting with Nova Scotia's efficient waste management practices. Transitioning to the United States, the data reveals historical trends, emphasizing the ongoing reliance on landfill disposal despite notable progress in recycling initiatives. China's waste management journey reflects rapid economic development, with per-person waste generation showing a more moderate increase compared to overall waste growth. Lastly, the waste composition data for Australia illustrates the significance of targeted initiatives to address the substantial quantities of organics, metals, and plastics. Collectively, these insights underscore the importance of sustained efforts to promote sustainable practices and circular waste management systems globally.

Reference:

Government of Canada, Statistics Canada. (2023, September 27). *Population estimates, quarterly*.

https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1710000901&request_locale=en

Government of Canada, Statistics Canada. (2023, May 2). *Disposal of waste, by source*.

<https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3810003201>

Environment and Climate Change Canada (2022). Reducing methane emissions from Canada's municipal solid waste landfills: discussion paper - Canada.ca. Canada.ca.

<https://www.canada.ca/en/environment-climate-change/services/canadian-environmental-protection-act-registry/reducing-methane-emissions-canada-municipal-solid-waste-landfills-discussion.html>

Statista. (2023, March 14). *Amount of disposed waste in China 1990-2021*.

<https://www.statista.com/statistics/279117/amount-of-disposed-garbage-in-china/>

China population 1950-2023. (n.d.). MacroTrends.

<https://www.macrotrends.net/countries/CHN/china/population>

U.S. population 1950-2023. (n.d.). MacroTrends.

<https://www.macrotrends.net/countries/USA/united-states/population>

National Overview: Facts and figures on materials, wastes and recycling | US EPA. (2022, December 3). US EPA.

<https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/national-overview-facts-and-figures-materials>

Waste Account, Australia, Experimental Estimates, 2018-19 financial year. (n.d.). Australian Bureau of Statistics.

<https://www.abs.gov.au/statistics/environment/environmental-management/waste-account-australia-experimental-estimates/latest-release>